## Statistical Properties of the OLS Coefficient Estimators

We have derived the OLS (Ordinary Least Squares) estimators of the regression coefficients in the simple linear regression model given by the **population regression equation**, or PRE

$$Y_i = \beta_0 + \beta_1 X_i + u_i \qquad (i = 1, \ldots, n) \qquad (1)$$

where $u_i$ is an iid random error term. The **OLS sample regression equation** (SRE) corresponding to PRE (1) is

$$Y_i = \hat{\beta}_0 + \hat{\beta}_1 X_i + \hat{u}_i \qquad (i = 1, \ldots, n) \qquad (2)$$

where $\hat{\beta}_0$ and $\hat{\beta}_1$ are the **OLS coefficient estimators** given by the formulas

$$\hat{\beta}_1 = \frac{\sum_i x_i y_i}{\sum_i x_i^2} \qquad (3)$$

$$\hat{\beta}_0 = \overline{Y} - \hat{\beta}_1 \overline{X} \qquad (4)$$

$x_i \equiv X_i - \overline{X}$, $y_i \equiv Y_i - \overline{Y}$, $\overline{X} = \sum_i X_i / n$ and $\overline{Y} = \sum_i Y_i / n$

## Why Use the OLS Coefficient Estimators?

The reason we use these OLS coefficient estimators is that, under assumptions A1-A8 of the classical linear regression model, they have several desirable statistical properties. We are going to examine these desirable statistical properties of the OLS coefficient estimators primarily in terms of the OLS slope coefficient estimator; the same properties apply to the intercept coefficient estimator.

- **PROPERTY 1: Linearity of $\hat{\beta}_1$**

The OLS coefficient estimator $\hat{\beta}_1$ can be written as a **linear function of the sample values of Y, the $Y_i$** $(i = 1, \ldots, n)$ .

**Proof:** Starts with formula (3) for $\hat{\beta}_1$:

$$\hat{\beta}_1 = \frac{\sum_i x_i y_i}{\sum_i x_i^2}$$

$$= \frac{\sum_i x_i (Y_i - \bar{Y})}{\sum_i x_i^2}$$

$$= \frac{\sum_i x_i Y_i}{\sum_i x_i^2} - \frac{\bar{Y} \sum_i x_i}{\sum_i x_i^2}$$

$$= \frac{\sum_i x_i Y_i}{\sum_i x_i^2} \qquad \text{because } \sum_i x_i = 0.$$

- Defining the observation weights $\mathbf{k_i} = \mathbf{x_i} / \Sigma_i \mathbf{x_i^2}$ for $(i = 1, \ldots, n)$ we can re-write the last expression above for $\hat{\beta}_1$ as:

$$\hat{\beta}_1 = \Sigma_i k_i Y_i \qquad \text{where } k_i \equiv \frac{x_i}{\Sigma_i x_i^2} \qquad (i = 1, \ldots, n) \qquad \ldots \textbf{(P1)}$$

- Note that the formula (3) and the definition of the weights $k_i$ imply that $\hat{\beta}_1$ is also a linear function of the $y_i$'s such that

$$\hat{\beta}_1 = \Sigma_i k_i y_i .$$

❑ ***Result:*** The OLS slope coefficient estimator $\hat{\beta}_1$ is a ***linear* function of the sample values $Y_i$ or $y_i$** $(i = 1, \ldots, n)$ where the coefficient of $Y_i$ or $y_i$ is $k_i$.

## Properties of the Weights $k_i$

In order to establish the remaining properties of $\hat{\beta}_1$, it is necessary to know the arithmetic properties of the weights $k_i$.

**[K1]** $\boldsymbol{\Sigma_i k_i = 0}$, i.e., the weights $k_i$ sum to zero.

$$\Sigma_i k_i = \Sigma_i \frac{x_i}{\Sigma_i x_i^2} = \frac{1}{\Sigma_i x_i^2} \Sigma_i x_i = 0, \qquad \text{because } \Sigma_i x_i = 0.$$

**[K2]** $\boldsymbol{\Sigma_i k_i^2 = \frac{1}{\Sigma_i x_i^2}}$.

$$\Sigma_i k_i^2 = \Sigma_i \left(\frac{x_i}{\Sigma_i x_i^2}\right)^2 = \Sigma_i \frac{x_i^2}{\left(\Sigma_i x_i^2\right)^2} = \frac{\left(\Sigma_i x_i^2\right)}{\left(\Sigma_i x_i^2\right)^2} = \frac{1}{\Sigma_i x_i^2}.$$

**[K3]** $\boldsymbol{\Sigma_i k_i x_i = \Sigma_i k_i X_i}$ .

$$\begin{aligned}\Sigma_i k_i x_i &= \Sigma_i k_i (X_i - \overline{X}) \\ &= \Sigma_i k_i X_i - \overline{X}\Sigma_i k_i \\ &= \Sigma_i k_i X_i \qquad\qquad \text{since } \Sigma_i k_i = 0 \text{ by [K1] above.}\end{aligned}$$

**[K4]** $\boldsymbol{\Sigma_i k_i x_i = 1}$ .

$$\Sigma_i k_i x_i = \Sigma_i \left(\frac{x_i}{\Sigma_i x_i^2}\right) x_i = \Sigma_i \frac{x_i^2}{\left(\Sigma_i x_i^2\right)} = \frac{\left(\Sigma_i x_i^2\right)}{\left(\Sigma_i x_i^2\right)} = 1 .$$

*Implication:* $\boldsymbol{\Sigma_i k_i X_i = 1.}$

- **PROPERTY 2:** **Unbiasedness of $\hat{\beta}_1$ and $\hat{\beta}_0$.**

The **OLS coefficient estimator** $\hat{\beta}_1$ **is unbiased**, meaning that $\mathbf{E}(\hat{\beta}_1) = \beta_1$.
The **OLS coefficient estimator** $\hat{\beta}_0$ **is unbiased**, meaning that $\mathbf{E}(\hat{\beta}_0) = \beta_0$.

- **Definition of unbiasedness:** The coefficient estimator $\hat{\beta}_1$ is unbiased if and only if $E(\hat{\beta}_1) = \beta_1$; i.e., its mean or expectation is equal to the true coefficient $\beta_1$.

- **Proof of unbiasedness of $\hat{\beta}_1$:** Start with the formula $\hat{\beta}_1 = \Sigma_i k_i Y_i$.

**1.** Since assumption A1 states that the PRE is $Y_i = \beta_0 + \beta_1 X_i + u_i$,

$$
\begin{aligned}
\hat{\beta}_1 &= \Sigma_i k_i Y_i \\
&= \Sigma_i k_i(\beta_0 + \beta_1 X_i + u_i) && \text{since } Y_i = \beta_0 + \beta_1 X_i + u_i \text{ by A1} \\
&= \beta_0 \Sigma_i k_i + \beta_1 \Sigma_i k_i X_i + \Sigma_i k_i u_i \\
&= \beta_1 + \Sigma_i k_i u_i, && \text{since } \Sigma_i k_i = 0 \text{ and } \Sigma_i k_i X_i = 1.
\end{aligned}
$$

**2.** Now take expectations of the above expression for $\hat{\beta}_1$, conditional on the sample values $\{X_i: i = 1, \ldots, n\}$ of the regressor X. Conditioning on the sample values of the regressor X means that the $k_i$ are treated as nonrandom, since the $k_i$ are functions only of the $X_i$.

$$
\begin{aligned}
E(\hat{\beta}_1) &= E(\beta_1) + E[\textstyle\sum_i k_i u_i] \\
&= \beta_1 + \textstyle\sum_i k_i E(u_i | X_i) && \text{since } \beta_1 \text{ is a constant and the } k_i \text{ are nonrandom} \\
&= \beta_1 + \textstyle\sum_i k_i 0 && \text{since } E(u_i | X_i) = 0 \text{ by assumption A2} \\
&= \beta_1.
\end{aligned}
$$

- ***<u>Result</u>:*** The **OLS slope coefficient estimator $\hat{\beta}_1$ is an *unbiased* estimator** of the slope coefficient $\beta_1$: that is,

$$\mathbf{E(\hat{\beta}_1) = \beta_1}. \qquad \textbf{... (P2)}$$

- **Proof of unbiasedness of $\hat{\beta}_0$:** Start with the formula $\hat{\beta}_0 = \overline{Y} - \hat{\beta}_1\overline{X}$.

**1.** Average the PRE $Y_i = \beta_0 + \beta_1 X_i + u_i$ across i:

$$\sum_{i=1}^{n} Y_i = n\,\beta_0 + \beta_1 \sum_{i=1}^{n} X_i + \sum_{i=1}^{n} u_i \qquad \text{(sum the PRE over the N observations)}$$

$$\frac{\sum_{i=1}^{n} Y_i}{n} = \frac{n\,\beta_0}{n} + \beta_1 \frac{\sum_{i=1}^{n} X_i}{n} + \frac{\sum_{i=1}^{n} u_i}{n} \qquad \text{(divide by n)}$$

$\overline{Y} = \beta_0 + \beta_1\overline{X} + \overline{u}$ where $\overline{Y} = \sum_i Y_i / n$, $\overline{X} = \sum_i X_i / n$, and $\overline{u} = \sum_i u_i / n$.

**2.** Substitute the above expression for $\overline{Y}$ into the formula $\hat{\beta}_0 = \overline{Y} - \hat{\beta}_1\overline{X}$:

$$\begin{aligned}
\hat{\beta}_0 &= \overline{Y} - \hat{\beta}_1\overline{X} \\
&= \beta_0 + \beta_1\overline{X} + \overline{u} - \hat{\beta}_1\overline{X} \qquad \text{since } \overline{Y} = \beta_0 + \beta_1\overline{X} + \overline{u} \\
&= \beta_0 + (\beta_1 - \hat{\beta}_1)\overline{X} + \overline{u}.
\end{aligned}$$

**3.** Now take the expectation of $\hat{\beta}_0$ conditional on the sample values $\{X_i: i = 1, \ldots, n\}$ of the regressor X. Conditioning <u>on</u> the $X_i$ means that $\overline{X}$ is treated as nonrandom in taking expectations, since X is a function only of the $X_i$.

$$
\begin{aligned}
E(\hat{\beta}_0) &= E(\beta_0) + E\left[(\beta_1 - \hat{\beta}_1)\overline{X}\right] + E(\overline{u}) \\
&= \beta_0 + \overline{X}\,E(\beta_1 - \hat{\beta}_1) + E(\overline{u}) && \text{since } \beta_0 \text{ is a constant} \\
&= \beta_0 + \overline{X}\,E(\beta_1 - \hat{\beta}_1) && \text{since } E(\overline{u}) = 0 \text{ by assumptions A2 and A5} \\
&= \beta_0 + \overline{X}\left[E(\beta_1) - E(\hat{\beta}_1)\right] \\
&= \beta_0 + \overline{X}\,(\beta_1 - \beta_1) && \text{since } E(\beta_1) = \beta_1 \text{ and } E(\hat{\beta}_1) = \beta_1 \\
&= \beta_0
\end{aligned}
$$

- ***<u>Result</u>:*** The **OLS intercept coefficient estimator** $\hat{\beta}_0$ **is an *unbiased* estimator** of the intercept coefficient $\beta_0$: that is,

$$\mathbf{E(\hat{\beta}_0) = \beta_0}\,. \qquad \textbf{... (P2)}$$

➢ **PROPERTY 3: Variance of $\hat{\beta}_1$ is minimum in the class of all linear and unbiased estimators.**

- **Definition:** The variance of the OLS slope coefficient estimator $\hat{\beta}_1$ is defined as

$$\operatorname{Var}(\hat{\beta}_1) \equiv E\left\{\left[\hat{\beta}_1 - E(\hat{\beta}_1)\right]^2\right\}.$$

- **Derivation of Expression for $\operatorname{Var}(\hat{\beta}_1)$:**

**1.** Since $\hat{\beta}_1$ is an unbiased estimator of $\beta_1$, $E(\hat{\beta}_1) = \beta_1$. The variance of $\hat{\beta}_1$ can therefore be written as

$$\operatorname{Var}(\hat{\beta}_1) = E\left\{\left[\hat{\beta}_1 - \beta_1\right]^2\right\}.$$

**2.** From part (1) of the unbiasedness proofs above, the term $[\hat{\beta}_1 - \beta_1]$, which is called the **sampling error of $\hat{\beta}_1$**, is given by

$$\left[\hat{\beta}_1 - \beta_1\right] = \Sigma_i k_i u_i .$$

**3.** The square of the sampling error is therefore

$$\left[\hat{\beta}_1 - \beta_1\right]^2 = \left(\sum_i k_i u_i\right)^2$$

**4.** Since the square of a sum is equal to the sum of the squares plus twice the sum of the cross products,

$$\begin{aligned}\left[\hat{\beta}_1 - \beta_1\right]^2 &= \left(\sum_i k_i u_i\right)^2 \\ &= \sum_{i=1}^{n} k_i^2 u_i^2 + 2\sum_{i<s}\sum_{s=2}^{n} k_i k_s u_i u_s .\end{aligned}$$

For example, if the summation involved only three terms, the square of the sum would be

$$\left(\sum_{i=1}^{3} k_i u_i\right)^2 = (k_1u_1 + k_2u_2 + k_3u_3)^2$$

$$= k_1^2u_1^2 + k_2^2u_2^2 + k_3^2u_3^2 + 2k_1k_2u_1u_2 + 2k_1k_3u_1u_3 + 2k_2k_3u_2u_3.$$

**5.** Now use assumptions A3 and A4 of the classical linear regression model (CLRM):

(A3) $\mathrm{Var}(u_i | X_i) = E(u_i^2 | X_i) = \sigma^2 > 0$ for all $(i = 1, \ldots, n)$

(A4) $\mathrm{Cov}(u_i, u_s | X_i, X_s) = E(u_i u_s | X_i, X_s) = 0$ for all $i \neq s$.

**6.** We take expectations conditional on the sample values of the regressor X:

$$E\left[\left(\hat{\beta}_1 - \beta_1\right)^2\right] = \sum_{i=1}^{n} k_i^2 E(u_i^2 \mid X_i) + 2\sum_{i<s}\sum_{s=2}^{n} k_i k_s E(u_i u_s \mid X_i, X_s)$$

$$= \sum_{i=1}^{n} k_i^2 E(u_i^2 \mid X_i) \quad \text{since } E(u_i u_s \mid X_i, X_s) = 0 \quad \text{for } i \neq s \text{ by (A4)}$$

$$= \sum_{i=1}^{n} k_i^2 \sigma^2 \quad \text{since } E(u_i^2 \mid X_i) = \sigma^2 \;\; \forall i \;\text{ by (A3)}$$

$$= \frac{\sigma^2}{\sum_i x_i^2} \quad \text{since } \sum_i k_i^2 = \frac{1}{\sum_i x_i^2} \text{ by (K2).}$$

- ***Result:*** The ***variance*** of the OLS slope coefficient estimator $\hat{\beta}_1$ is

$$\operatorname{Var}(\hat{\beta}_1) = \frac{\sigma^2}{\sum_i x_i^2} = \frac{\sigma^2}{\sum_i (X_i - \bar{X})^2} = \frac{\sigma^2}{TSS_X} \quad \text{where } TSS_X = \sum_i x_i^2 . \quad \textbf{... (P3)}$$

$$\hat{\beta}_1 = \sum k_i Y_i$$

where

$$k_i = \frac{X_i - \bar{X}}{\sum (X_i - \bar{X})^2} = \frac{x_i}{\sum x_i^2} \qquad \text{(see Appendix 3A.2)}$$

which shows that $\beta_2$ is a weighted average of the $Y$'s, with $k_i$ serving as the weights.

Let us define an alternative linear estimator of $\beta_2$ as follows:

$$\beta_1{}^* = \sum w_i Y_i$$

where $w_i$ are also weights, not necessarily equal to $k_i$. Now

$$\begin{aligned} E(\beta_1{}^*) &= \sum w_i E(Y_i) \\ &= \sum w_i (\beta_0 + \beta_1 X_i) \\ &= \beta_0 \sum w_i + \beta_1 \sum w_i X_i \end{aligned}$$

Therefore, for $\beta_2^*$ to be unbiased, we must have

$$\sum w_i = 0$$

and

$$\sum w_i X_i = 1$$

Also, we may write

$$
\begin{aligned}
\operatorname{var}(\beta_1{}^*) &= \operatorname{var} \sum w_i Y_i \\
&= \sum w_i^2 \operatorname{var} Y_i \qquad [Note\!: \operatorname{var} Y_i = \operatorname{var} u_i = \sigma^2] \\
&= \sigma^2 \sum w_i^2 \qquad [Note\!: \operatorname{cov}(Y_i, Y_j) = 0 \, (i \neq j)] \\
&= \sigma^2 \sum \left( w_i - \frac{x_i}{\sum x_i^2} + \frac{x_i}{\sum x_i^2} \right)^2 \qquad \text{(Note the mathematical trick)} \\
&= \sigma^2 \sum \left( w_i - \frac{x_i}{\sum x_i^2} \right)^2 + \sigma^2 \frac{\sum x_i^2}{\left(\sum x_i^2\right)^2} + 2\sigma^2 \sum \left( w_i - \frac{x_i}{\sum x_i^2} \right) \left( \frac{x_i}{\sum x_i^2} \right) \\
&= \sigma^2 \sum \left( w_i - \frac{x_i}{\sum x_i^2} \right)^2 + \sigma^2 \left( \frac{1}{\sum x_i^2} \right)
\end{aligned}
$$

because the last term in the next to the last step drops out. (Why?)

Since the last term in above equation is constant, the variance of $(\beta_2{}^*)$ can be minimized only by manipulating the first term. If we let

$$w_i = \frac{x_i}{\sum x_i^2}$$

$$
\begin{aligned}
\operatorname{var}(\beta_1{}^*) &= \frac{\sigma^2}{\sum x_i^2} \\
&= \operatorname{var}(\hat{\beta}_2)
\end{aligned}
$$

In words, with weights $w_i = k_i$ , which are the least-squares weights, the variance of the linear estimator $\beta_1^*$ is equal to the variance of the least-squares estimator $\widehat{\beta_1}$; otherwise $\text{var}(\beta_1^*) > \text{var}(\hat{\beta}_1)$. To put it differently, if there is a minimum-variance linear unbiased estimator of $\beta_1$, it must be the least squares estimator of $\beta_1$.